

Linux running notes

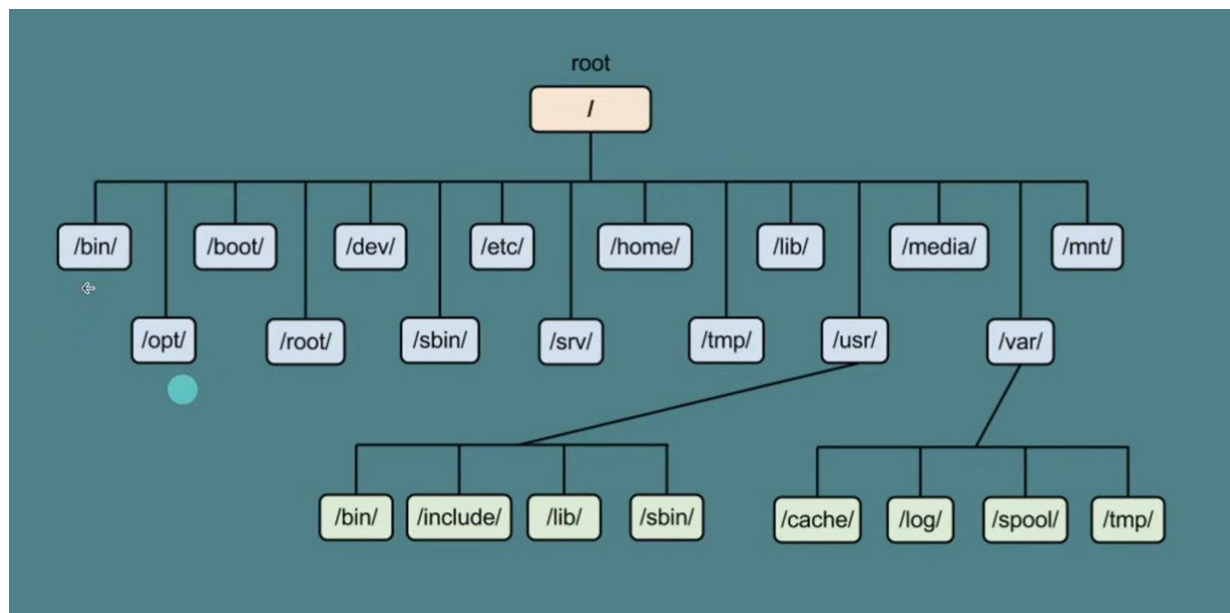
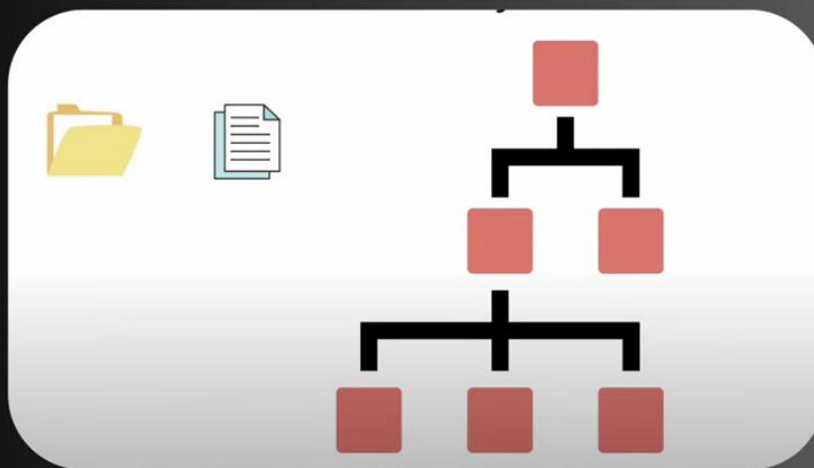
18/04/26

The File System

OS store and manage data on disk or Partitions using a structure called **Filesystem**.

File system includes files, directories and it's related permissions.

It stores data in hierarchy of directories and files.



It is Case Sensitive and Extension

There is no extension and file type in Linux

>> touch file , echo hello > file , cat file

➤ Type of File System

ext3(Extended 3)

ext4

XFS (Extended File System):

-(Ubuntu, Debian, Fedora, CentOS, RedHat)

BtrFS(B-tree FS)

(OpenSUSE and SUSE Linux Enterprise Server)

FAT(File allocation table)

➤ How to check your Linux Filesystem?

lsblk -f

df -Th

cat /etc /fstab

XFS vs EXT4

- **XFS**

- Optimized for large files and volumes, offering superior performance and scalability.

- **EXT4**

- Performs well across various file sizes but less efficient with extremely large files.

For High-throughput and extensive parallel processing capabilities, XFS is typically more effective than ext4.

Core Components of the Linux Environment

Before starting with shell scripting, it is important to understand the main components of a Linux system. These components work together in layers, where user input passes through multiple stages before reaching the hardware.

- **Terminal:** Interface used to access the shell
- **Shell:** Command interpreter that processes user input
- **Kernel:** Core of the operating system that manages hardware and system resources

- **GNU Utilities & Libraries:** Provide essential commands and system functions

Complete Linux system = Kernel + GNU system utilities and libraries + other management scripts + installation scripts.

➤ **How to check your Linux filesystem?**

- lsblk -f
- df -Th
- cat/ etc/fstab

```
parallels@ubuntu01: /
parallels@ubuntu01:~$
parallels@ubuntu01:~$
parallels@ubuntu01:~$ cd /
parallels@ubuntu01:/$ ls
bin  dev  home  lost+found  mnt  proc  run  snap  swap.img  tmp  var
boot  etc  lib  media      opt  root  sbin  srv   sys       usr
parallels@ubuntu01:/$
parallels@ubuntu01:/$ df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
tmpfs           tmpfs     197M   1.9M  195M   1% /run
/dev/sda2       ext4      62G   12G   47G  21% /
tmpfs           tmpfs     984M    0   984M   0% /dev/shm
tmpfs           tmpfs     5.0M   4.0K   5.0M   1% /run/lock
/dev/sda1       vfat      1.1G   6.4M   1.1G   1% /boot/efi
tmpfs           tmpfs     197M  128K   197M   1% /run/user/1000
iCloud          prl_fs    927G  739G   189G  80% /media/psf/iCloud
parallels@ubuntu01:/$
```

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What is inode?

An inode in Linux is a data structure that stores metadata about a file or directory.

The filesystem uses the inode number to locate the inode, which then contains pointers to the data blocks where the actual file data is stored.

➤ What is inode?

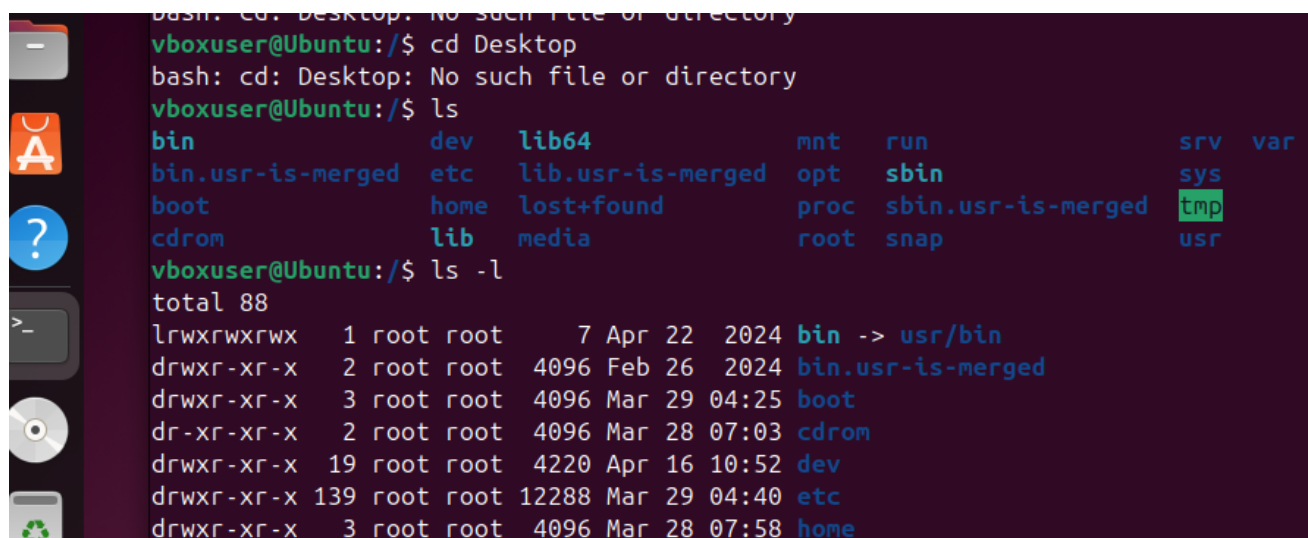
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➤ Basic New commands

- mv : to rename file
- ls -l : to know the full detail
- ls -li : It also give inod number

➤ /(root) Directory: The Linux filesystem starts at the root directory, denoted by a single slash (/), from which all other files and directories branch out.



```
bash: cd: Desktop: No such file or directory
vboxuser@Ubuntu:/$ cd Desktop
bash: cd: Desktop: No such file or directory
vboxuser@Ubuntu:/$ ls
bin          dev          lib64        mnt          run          srv          var
bin.usr-is-merged  etc          lib.usr-is-merged  opt          sbin         sys
boot         home         lost+found    proc         sbin.usr-is-merged tmp
cdrom        lib          media        root         snap         usr
vboxuser@Ubuntu:/$ ls -l
total 88
lrwxrwxrwx   1 root root    7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   3 root root 4096 Mar 29 04:25 boot
dr-xr-xr-x   2 root root 4096 Mar 28 07:03 cdrom
drwxr-xr-x  19 root root 4220 Apr 16 10:52 dev
drwxr-xr-x 139 root root 12288 Mar 29 04:40 etc
drwxr-xr-x   3 root root 4096 Mar 28 07:58 home
```

- /bin : Contains essential user binaries(executable), such as common commands like ls, cp, etc.
- /etc: Holds system configuration files. Ex. User, Network, Services, System apps
- /home: Contains the personal directory of all users.
- /root: the home directory for the root user (administrator).
- /var: where variables data such as logs and databases are stored.
- /tmp: Temporary files created by system and users.

```
vboxuser@Ubuntu:~/Desktop/LnxPrac$ ./NewPrac2.sh
Hey this is the System performance info
+ top -es
top: bad memory scaling arg 's'
+ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            536M  2.1M  534M   1% /run
/dev/sda2        25G   7.7G   16G  34% /
tmpfs            2.7G    0  2.7G   0% /dev/shm
tmpfs            5.0M   8.0K  5.0M   1% /run/lock
tmpfs            536M  128K  535M   1% /run/user/1000
/dev/sr0         51M   51M    0 100% /media/vboxuser/VBox_GAs_7.2.6
+ free -g
              total        used        free      shared  buff/cache   available
Mem:           5             1             2             0             1             4
Swap:          0             0             0
+ nproc
4
```

```
vboxuser@Ubuntu: ~/Desktop/LnxPrac
vboxuser@Ubuntu:~/Desktop/LnxPrac$ ps -ef | grep firefox
vboxuser      5084      1996   4 09:41 ?           00:01:11 /snap/firefox/7766/usr/lib/firefox/f
firefox
vboxuser      5164      1762   0 09:41 ?           00:00:00 /snap/firefox/7766/usr/lib/firefox/c
rashhelper 5084 9 /tmp/ 11
vboxuser      5251      5084   0 09:41 ?           00:00:00 /snap/firefox/7766/usr/lib/firefox/f
firefox -contentproc -ipcHandle 0 -signalPipe 1 -initialChannelId {3bfa9be6-8586-437c-8db
8-8758262106da} -parentPid 5084 -greomni /snap/firefox/7766/usr/lib/firefox/omni.js -app
omni /snap/firefox/7766/usr/lib/firefox/browser/omni.js -appDir /snap/firefox/7766/usr/l
ib/firefox/browser 1 forkserver
vboxuser      5255      5251   0 09:41 ?           00:00:00 /snap/firefox/7766/usr/lib/firefox/f
firefox -contentproc -parentBuildID 20260204004611 -prefsHandle 0:35513 -prefMapHandle 1:
276825 -sandboxReporter 2 -chrootClient 3 -ipcHandle 4 -initialChannelId {d953f261-09b9-
4464-9dee-b9bc9d8b4e11} -parentPid 5084 -crashReporter 5 -crashHelper 6 -appDir /snap/fi
refox/7766/usr/lib/firefox/browser 2 socket
vboxuser      5285      5251   0 09:41 ?           00:00:05 /snap/firefox/7766/usr/lib/firefox/f
firefox -contentproc -isForBrowser -prefsHandle 0:35669 -prefMapHandle 1:276825 -jsInitHa
ndle 2:223968 -parentBuildID 20260204004611 -sandboxReporter 3 -chrootClient 4 -ipcHandl
e 5 -initialChannelId {46d9bac4-9811-4edc-8e33-5bc5c87e8e24} -parentPid 5084 -crashRepor
ter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.js -appomni /snap/
firefox/7766/usr/lib/firefox/browser/omni.js -appDir /snap/firefox/7766/usr/lib/firefox/
browser 3 tab
vboxuser      5295      5251   0 09:41 ?           00:00:00 /snap/firefox/7766/usr/lib/firefox/f
firefox -contentproc -parentBuildID 20260204004611 -prefsHandle 0:35669 -prefMapHandle 1:
276825 -sandboxReporter 2 -chrootClient 3 -ipcHandle 4 -initialChannelId {246c2465-2976-
```